Part A — Theoretical Foundation

Derivable Judgement: A Statistical Decision-Making Model

1. What is inferential statistics?

Inferential statistics is the branch of statistics concerned with drawing conclusions about a population from a limited sample of it. Rather than measuring every individual, we measure a representative subset and use probability theory to estimate how much the sample result would vary if we drew a different sample — which lets us say how confidently the pattern likely holds for the full population.

2. What is hypothesis testing and its components?

Hypothesis testing is a structured procedure for deciding between two competing claims about a population using sample evidence. Its core components are: the null hypothesis (H0), a default claim of no effect or no difference; the alternative hypothesis (H1), the claim we adopt if the evidence contradicts H0; a test statistic computed from the sample; a significance level (alpha, commonly 0.05) fixing how much risk of a wrong rejection we'll tolerate; and a decision rule that compares the p-value or test statistic against that threshold to accept or reject H0.

3. Explain confidence interval and critical value.

A confidence interval is a range built around a sample statistic (such as a mean) that is expected to contain the true population parameter a stated percentage of the time (e.g., 95%) if the sampling process were repeated many times. It communicates the precision of an estimate, not just a single point guess. A critical value is the cutoff point on a test statistic's distribution that separates 'consistent with H0' from 'unlikely enough to reject H0,' set by the chosen significance level; if the observed test statistic falls beyond it, the result is declared statistically significant.

4. Define p-value.

The p-value is the probability of observing a result at least as extreme as the one actually obtained, assuming the null hypothesis is true. A small p-value (typically below the chosen alpha, e.g. 0.05) indicates the observed data would be unusual under H0, giving grounds to reject it; it is not the probability that H0 itself is true.

5. Differentiate Type I and Type II errors.

A Type I error occurs when a true null hypothesis is incorrectly rejected — a false positive, with probability controlled by alpha. A Type II error occurs when a false null hypothesis is incorrectly retained — a false negative, with probability denoted beta. There is a built-in trade-off between the two: tightening alpha to reduce false positives generally raises the chance of missing a real effect (higher beta), so the acceptable balance depends on which mistake is costlier in context.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

Z-test: compares a sample mean (or proportion) to a known value, or compares two means, when the population standard deviation is known and/or the sample is large (conventionally $n > 30$), relying on the normal distribution. **T-test**: compares one or two group means when the population standard deviation is unknown and must be estimated from the sample, which is the more common real-world situation, especially with smaller samples; it uses the t-distribution to account for that extra uncertainty. **Chi-square test**: assesses whether two categorical variables are associated (test of independence) or whether an observed categorical distribution matches an expected one (goodness-of-fit), by comparing observed versus expected counts in a contingency table. **ANOVA (Analysis of Variance)**: compares the means of three or more independent groups simultaneously to test whether at least one group's mean differs from the others, avoiding the inflated false-positive rate that would come from running many separate two-group tests.

7. What is Covariance?

Covariance measures the direction in which two variables move together. A positive covariance means that when one variable is above its own mean, the other tends to be above its mean too (they move in the same direction); a negative covariance means they tend to move in opposite directions. Because its magnitude depends on the units and scale of the variables, covariance is useful for direction but not for judging how strong a relationship is.

8. What is Correlation?

Correlation standardizes covariance into a unitless value between -1 and +1, making it possible to judge both the direction and the strength of a linear relationship, and to compare relationships across different variable pairs. A value near +1 indicates a strong positive linear relationship, near -1 a strong negative one, and near 0 indicates little to no linear relationship. Correlation, like covariance, describes association and does not by itself establish that one variable causes changes in the other.